
Making Web Bluetooth Work
A field report from a chip on the desk
Ifedayo Agboola · XtremeJS 2026

What we're building today

A web page that talks to hardware, with nothing to install.

Chip on the Desk

Web Bluetooth, Chrome, one NXP QN9090 running the Proximity Reporter firmware

● Chooser cancelled

1. CONNECTION

Find the chip | Disconnect | Forget

Device name	–
Device id (per origin)	–
Manufacturer	–
Model	–
Firmware	–
Hardware	–
TX power	–
Permission state	query unsupported

The chooser only shows devices advertising the Immediate Alert service (0x1802). One device per pick, a user gesture every time, HTTPS or localhost only.

2. IMMEDIATE ALERT (WRITE WITHOUT RESPONSE)

No alert | Mild alert | High alert

Writes one byte to characteristic 0x2A06. On the DK6 board: mild turns LED2 on, high makes LED2 flash, no alert turns it off.

LINK LOSS LEVEL (WRITE WITH RESPONSE)

Off | Mild | High current: –

The chip keeps this. When the link drops, the chip itself raises the alert: mild flashes the blue LED, high flashes the green one. The browser only finds out through `gattserverdisconnected`.

3. BATTERY (READ + NOTIFY)

— Read once

The firmware reports a fixed 90%. The point is the shape: one read, then `startNotifications` and a `characteristicvaluechanged` listener.

4. THE GATT QUEUE

☑ Route operations through the queue

Fire 8 reads at once

Unqueued, concurrent GATT calls race the one radio link and Chrome rejects with `NetworkError: GATT operation already in progress`. Queued, they run one after another and all succeed.

5. RECONNECT POLICY

☑ Reconnect after an unexpected drop

Attempts	0
Next delay	–

6. RSSI

— Try watchAdvertisements()

After `connect()` Web Bluetooth gives no signal strength at all.

A tag that shouted, and a phone that listened.

The bike tracker: no connection, no pairing, twenty bikes at once.


PACIFIC HEIGHTS
CHINATOWN
UNION SQUARE
San Francisco
MISSION DISTRICT
NOE VALLEY
Google
Assets
Lauren's bike
Near Belfast city centre Since 12:30pm
Company truck
Near Belfast city centre Since 12:30pm
History
Location
Profile
pnpm-lock.yaml
README.md
tailwind.config.js
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
Uploading assets skipped - no new assets found
69 iOS assets, 69 Android assets (maximum: 200
Published!
Branch
Runtime version
Platform
Update group ID
Android update ID
iOS update ID
Message
Commit
Website link
preview
1.0.1
android, ios
https://expo.dev/accounts/ifedayo
mesh/updates/d2d12300-83ee-43
https://expo.dev/accounts/ifedayo
OUTLINE
TIMELINE
bike-mesh git:(main) git
main* (Rebasing)
Launchpad
esc
tab
caps lock
Q
W
E
R
A
S
D

What makes this different

This time the listener is a browser tab.

And the chip had to change too.

Meet the board

A radio and a small computer on one chip.

NXP QN9090 on the DK6 kit: buttons, four lamps, a USB port.

OM15076-3 / JN5189
Carrier Main Board
DK6 V3
NXP
Generic Expansion Board E3378G2 OM15082-2
SW1
SW2
SW3
SW4
USER INTERFACE
TEST
PASS
FTDI USB
PCB2455-3

A board does nothing until you give it a program

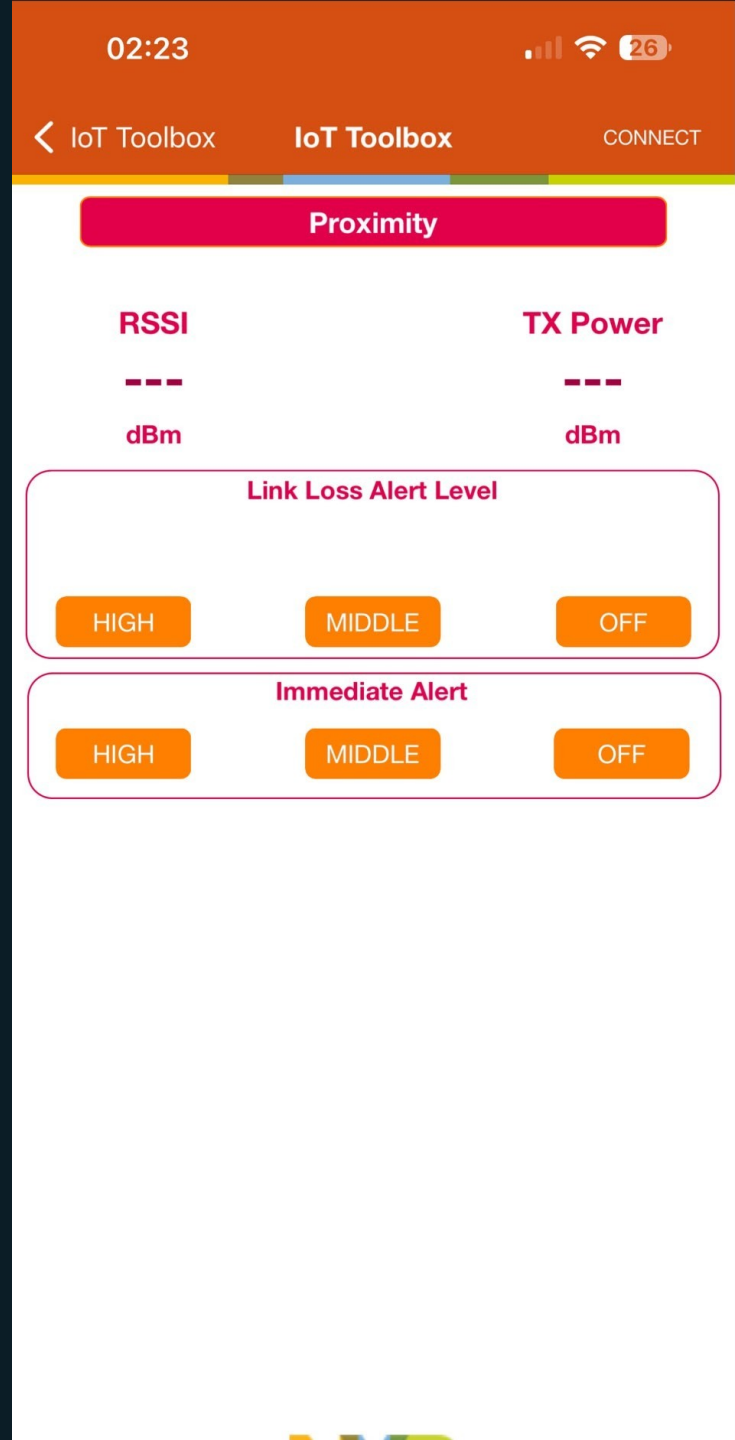
Firmware decides what a browser can see.

Program one shouts. Program two converses. Browsers only converse.

What the board can say now

A conversation is a menu.

Sections, items, and a few bytes for each one.



Web Bluetooth turns "download our app" into "open this link".

Who already uses it

LEGO Education

Home Assistant

Bangle.js

Ledger

Niimbot

Real products ship on it today.

Set up, control, install, sign, print.


9:41
Approve
Press both buttons
to validate transaction
Amount to swap
Amount to receive
Approve
LEDGER

The user picks the device, every time.

One click, one device, HTTPS only.

Seven lines of JavaScript

```
const device = await navigator.bluetooth.requestDevice({
  filters: [{ services: ['immediate_alert'] }],
  optionalServices: ['link_loss', 'tx_power', 'battery_service']
});
const server  = await device.gatt.connect();
const service = await server.getPrimaryService('immediate_alert');
const alert   = await service.getCharacteristic('alert_level');
await alert.writeValueWithoutResponse(Uint8Array.of(2));
```

Seven honest lines.

Ask, connect, open the section, pick the item, send the bytes.

Demo

A byte from a tab lights a lamp.

Demo one: tab to lamp

```
NetworkError: GATT operation already in progress
```

The radio answers one question at a time.

Chrome refuses the second. Open issue since 2015.

The queue

```
class GattQueue {
  tail = Promise.resolve();
  run(op) {
    const next = this.tail.then(op, op);
    this.tail = next.catch(() => {});   // a failure never breaks the chain
    return next;
  }
}
const gatt = new GattQueue();
await gatt.run(() => battery.readValue());
```

Take turns in your own code.

One line; every operation waits its turn.

Links drop late and silently.

Reopen the menu, time out every connect, back off, then stop.

Demo

Kill the link and watch it come back.

Demo two: kill the line

The number the browser won't give you

−65 −72 −78 −70 −85 −68 −92 −74 −69 −88

The browser will not give you this number.

Signal strength stays on the phone.

Where it works, November 2026



Chrome



Edge



Samsung Internet



Firefox



Safari

Most of the world's browsers, and none of Apple's phones.

What to build, and how to build it properly

Build it for the device next to the person.

A check, a queue, a timeout, a reconnect plan, a log, no secrets.


VIOTEK
SPECTRE

The path from chip to tab keeps getting shorter.

Two engines in, two out, and a scanning spec still alive.

What happened this morning



A lamp on the desk moved from a tab.

Thank you.

Demo page and slides:

github.com/ifedayoagboola/chip-on-the-desk

The tracker write-up:

dev.to/blackscripts

linkedin.com/in/ifedayoagboola



demo page and slides

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